AI Report 1 Learning with AI Step by Step

CSC6410 | 1 November 2026

This report describes how I used AI for Experiment 1 and checked its advice against Windows traceroutes and RIPEstat. I completed Homework 1 independently and did not use AI for its seven answers, so I do not claim any homework calculation checks.

1 Tasks and starting point

Homework 1 was submitted on 29 October 2026 at 12:08. I later asked AI what the assignment instructions meant, but did not use it to solve the homework. For Experiment 1, I knew I needed to trace five sites, map public hops to AS numbers, and compare with BGP. I was confused about how the three assignments fit together and the order of steps.

The earlier AI discussion happened before any five-target prediction table. At 12:43:58 IST on 1 November, an AI-assisted estimate table was saved before the first traceroute. I cannot describe that as unaided pre-consultation work. I used Windows on my home network.

2 Step by step AI log

#	What I asked / task	AI answer	What I checked	What happened / learned
1	Homework instructions: what should I ask AI about?	AI said homework AI use should support understanding, not replace my steps.	I had already submitted Homework 1 and confirmed I had not used AI for its answers.	There is no earlier homework calculation exchange to log.
2	I pasted Experiment 1 and asked for a clear explanation.	AI separated the experiment from the AI learning report and listed the five targets, two traces, mapper, and BGP comparison.	I checked this against the assignment, including the one-hour gap and separate .py file.	The experiment supplies the data; the AI report records how help was used.
3	I asked for a Windows walkthrough starting with predictions.	AI said to write my own estimates before commands.	No student-written prediction page was supplied; the later table was time stamped before measurement but after AI advice.	I cannot relabel AI-assisted estimates as independent predictions.
4	I asked Codex to run the experiment end to end on my Windows machine.	AI drafted a cached RIPEstat mapper and ran two trace sets more than an hour apart.	I have raw output, start times, the script, and a RIPE response. The script was rerun after a real error.	The data came from my machine; the explanations and draft code are AI-assisted.
5	Mapping problem: public hops had no ASN.	AI suggested one bgp-state fallback and leaving the hop unknown if no route appeared.	Fallback returned zero routes for CUW hop 11 and Wikipedia hop 13.	An unmapped interface is a data limit, not evidence that no AS carried the packet.
6	Privacy-mask bug: no intermediate public hop on some traces.	AI added a presence check before splitting the IP into octets.	A rerun raised AttributeError on NoneType; the next run completed all five sites.	I should test traces where only the destination is public.

Rows 5 and 6 were follow-up work under my end-to-end request, not separately typed prompts.

3 Where AI was wrong vague or incomplete

The first suggested privacy method treated the first visible public address as my home public IP. For CUW and Oxford, the first visible public IP was the destination itself, so this masked useful evidence. After changing the rule, the code briefly crashed when no public intermediate hop existed. The final code checks for that case. AI could not turn a later AI-assisted estimate into a genuine pre-consultation prediction.

4 What I learned

A tracert hop is one responding router address, while an AS is a network in interdomain routing. Several IP hops can belong to one AS, and many routers may not answer. Wikipedia's mapped trace showed Reliance Jio AS64049 followed by Wikimedia AS14907. CUW's web address resolved to Amazon AS16509, showing why I need to check the actual destination IP instead of assuming the institution's ASN. This is an AI-assisted draft of my understanding that I checked against the measurements.

5 Independent verification

I used the command output from my Windows computer. Run 1 reached all five destinations. CUW showed 17 IP hops and 8 timeout rows; run 2 showed 17 IP hops and 8 timeout rows. The RIPEstat browser JSON for Wikipedia's destination returned AS14907 and prefix 103.102.166.0/24. I checked RIPE NCC documentation for the endpoint meanings.

The mapped path is partial because private hops are skipped, some time out, and two public interfaces had no ASN. I compared the last visible ASN with the destination origin and three collector paths for each site. The collectors start from other locations and need not include the ASes visible near my home.

6 Looking back

Next time I would save my own predictions with a timestamp before opening an AI chat, ask narrower questions when a real output or error appears, and capture the screen at that moment. I would still verify AS claims against raw traces and RIPEstat rather than accept a plausible-sounding answer.

A separate PDF contains the relevant AI message transcript. Windows usernames and personal names in file paths were redacted for privacy at my request, so that copy is not literally unedited. I did not have earlier homework problem-solving AI conversations because I completed Homework 1 myself. The transcript should be compared with the chat's own Print or Export command before submission because this conversation was active while the report was prepared.